

INSTALLATION GUIDE

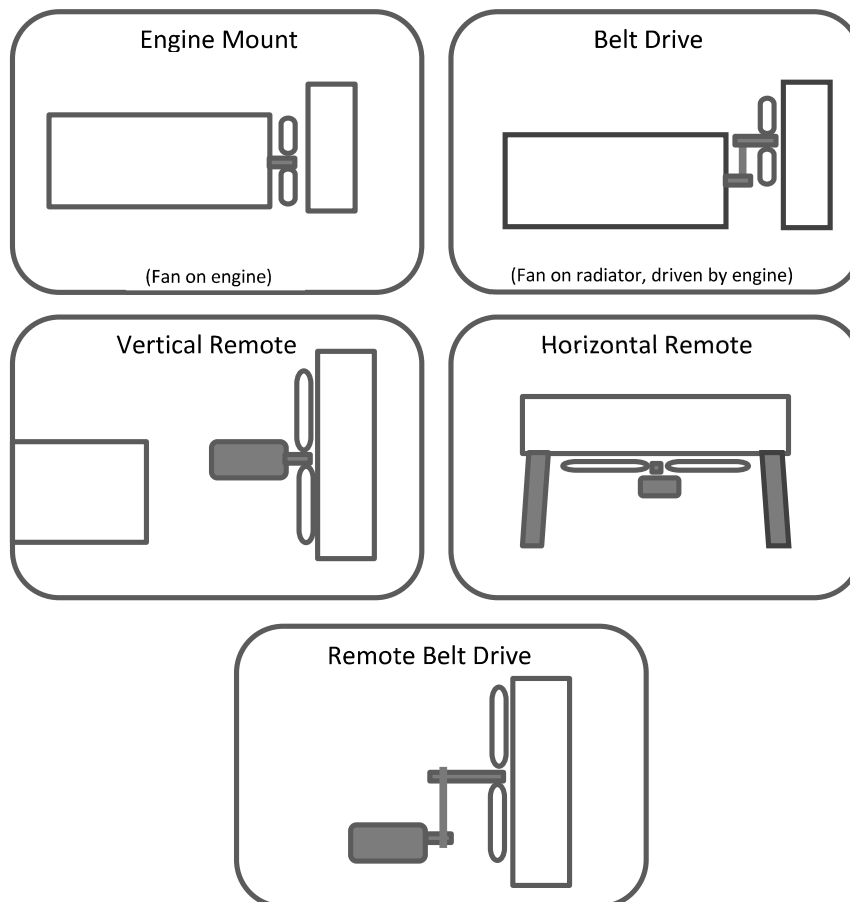
Always follow engine manufacturer's installation instructions.

- This guide gives information on general site installations for all radiator types and should be followed to achieve optimum performance and service life of the radiator.
- Specific instructions will be provided for special installations.
- Read and fully understand this guide and any other supplied documentation such as installation drawings, handling instructions etc. before attempting to install the radiator.
- For more detail on any of the points below please contact Bearward Engineering Technical Sales on +44 (0)1604 762851 -

IF IN DOUBT – ASK

1. Engine / Radiator configurations:

The schematics below identify the types of radiator referred to in this guide.



Check the type of radiator to be installed against the list below and ensure the installation satisfies the relevant points.

2. Positioning and Mounting

	Radiator Type				
	Engine Mount	Belt Drive	Vertical I Remote	Horizontal Remote	Remote Belt Drive
Make sure the altitude and ambient temperature that the generator will be operating in is within the specifications provided	⊙	⊙	⊙	⊙	⊙
Lift using only recommended lifting points detailed on radiator	⊙	⊙	⊙	⊙	⊙
Do not use radiator lifting points to lift engine / generator assembly	⊙	⊙	⊙	⊙	⊙
Position radiator on solid level ground, not on loose soil or vegetation			⊙	⊙	⊙
Position generator set on solid level ground, not on loose soil or vegetation	⊙	⊙			
Trailer mounted generators: Position on level ground, on correctly inflated tyres. The trailer must be in sound condition and have the correct fixing to the container. The ISO container must not have additional holes cut as this will cause a weakness in the structure and allow deformation.	⊙	⊙			
Allow a clear space of 2x maximum radiator dimensions in front of exit louvers to allow adequate air flow.	⊙	⊙	⊙		⊙
Allow a 1x maximum radiator dimension all around or a 1.5 x maximum dimension on three sides to allow adequate air for engine combustion, engine cooling and alternator cooling.				⊙	
Container or canopy installations – Make sure there is space to allow for adequate air to enter and exit the enclosure.	⊙	⊙			
Horizontal radiators should not be close coupled unless specifically designed as such.				⊙	
Make sure the radiator is square to the engine in all 3 Axes	⊙	⊙			
Do not hang non-specified or non-Bearward supplied supports or equipment from radiator or pipework	⊙	⊙	⊙	⊙	⊙
Mount engine / generator base as recommended by engine manufacturer. Radiator must be rigidly mounted same base as engine.	⊙	⊙			
Fans must be positioned correctly within the radiator cowl as per Bearward installation drawing.	⊙	⊙			⊙
Flexible ducting must be used to connect to air outlet of container, building or ship.	⊙	⊙	⊙	⊙	⊙
Make sure radiator bracing, where included in design, is attached to the correct holes in the radiator steelwork, to prevent backwards and forwards movement.	⊙	⊙	⊙	⊙	⊙
For aqueous marine environment (coastal / marine) / Tropical / High air pollution applications, ask Bearward Engineering for recommended products.	⊙	⊙	⊙	⊙	⊙
Appropriate cleaning – instruction can be provided.	⊙	⊙	⊙	⊙	⊙
Hot air / exhaust recirculation to be avoided.	⊙	⊙	⊙	⊙	⊙
Pipework lagging should be only done on engine manufacturers recommendations	⊙	⊙	⊙	⊙	⊙